

SΔφ-15

Trust, Hope, and Promise as Pre-Commitment Operators

Minimal Relational Persistence after SΔφ-14 (v1.0)

Version: v1.0

Series: Sofience-Δφ Formalism (SΔφ)

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This paper formalizes trust, hope, and promise as minimal operators that insert unobserved future transitions into present evaluation. It follows SΔφ-14, where ethics was defined as discrimination between real and phantom shared attractors. SΔφ-15 fixes the pre-commitment layer that makes such validation operable in time.

Series references

Items below are referenced for continuity and navigation within the SΔφ series.

Item	Title	DOI
SΔφ-00	The Sofience-Δφ Formalism: Conceptual Architecture and Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Sofience-Δφ Minimal Core: Phase-Change Formalism of Existence (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject as Interpretive Emergence: Extension of the Sofience-Δφ Minimal Core (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility as the Minimal Condition of Existence (v1.1)	10.5281/zenodo.18730448
SΔφ-04	Minimal Operational Conditions for AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency as Recursive Transition Law Update: Minimal Axiomatic Formulation (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility as Irreversible Cost Internalization: Minimal Axiomatic Formulation (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min — Interpretation Minimal Axioms (v1.0)	10.5281/zenodo.18735006
SΔφ-08	Agency-Responsibility-Freedom Closure: Minimal Axiomatic Formulation (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error, Boundary, and Freedom: Emergent Limits within the ARF Structure (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation, Learning, and Inference as a Single Re-entry Structure (v1.0)	10.5281/zenodo.18746509

Item	Title	DOI
SΔφ-11	World, Other, Self, and Consciousness as Boundary Phenomena (v1.0)	10.5281/zenodo.18753403
SΔφ-12	Memory and Forgetting as Irreversible Trace Regulation (v1.0)	10.5281/zenodo.18755685
SΔφ-13	Meaning, Value, and Pain as Emergent Phenomena (v1.0)	10.5281/zenodo.18759458
SΔφ-14	Love, Sacrifice, and Ethics as Attractor Coupling: Minimal Relational Dynamics after Meaning-Value-Pain (v1.0)	10.5281/zenodo.18761928

0. Position statement

$S\Delta\phi$ -14 defined ethics as discrimination between real and phantom shared attractors under irreversible coupling. Validation is structurally delayed: it happens after commitment. This item fixes pre-commitment operators that keep relational persistence open under uncertainty.

1. Axiom 15-1 — Trust

Trust is a pre-commitment to unobserved future transitions of the other.

Trust := Insert($\Delta\phi_{\text{other}}^{\text{future}}$) into Eval_self(t).

Failure mode: trust failure = betrayal = collapse of relational persistence.

2. Axiom 15-2 — Belief

Belief is a stabilized hypothesis about world-transition rules.

Belief := Fix(H($\Delta\phi_{\text{world}}$)).

Failure mode: belief failure = illusion = persistent mismatch with world $\Delta\phi$.

3. Lemma 15-A — Trust $\neq$ Belief

Aspect	Trust	Belief
Target	Other / relation	World / model
Function	Opens a relational path	Stabilizes a world hypothesis
Primary risk	Betrayal (relation collapse)	Illusion (model mismatch)
Temporal direction	Future-other $\rightarrow$ present evaluation	Past pattern $\rightarrow$ fixed rule

4. Axiom 15-3 — Hope

Hope is suspension of closure: the system refrains from declaring Ω_{future} closed.

Hope := Suspend_Closure(Ω_{future}). Despair := Close(Ω_{future}).

5. Axiom 15-4 — Promise

Promise is distributed commitment: SharedCommit($\Delta\phi_{\text{future}}$), requiring ≥ 2 systems.

Promise := SharedCommit($\Delta\phi_{\text{future}}$).

Break(Promise) $\Rightarrow$ Collapse(A_shared).

6. Proposition 15-P — Temporal inversion of responsibility

With promise, unobserved futures constrain present actions: future constraints shape current choices.

FutureConstraint $\rightarrow$ PresentActionRestriction (responsibility reversal).

7. Boundary note

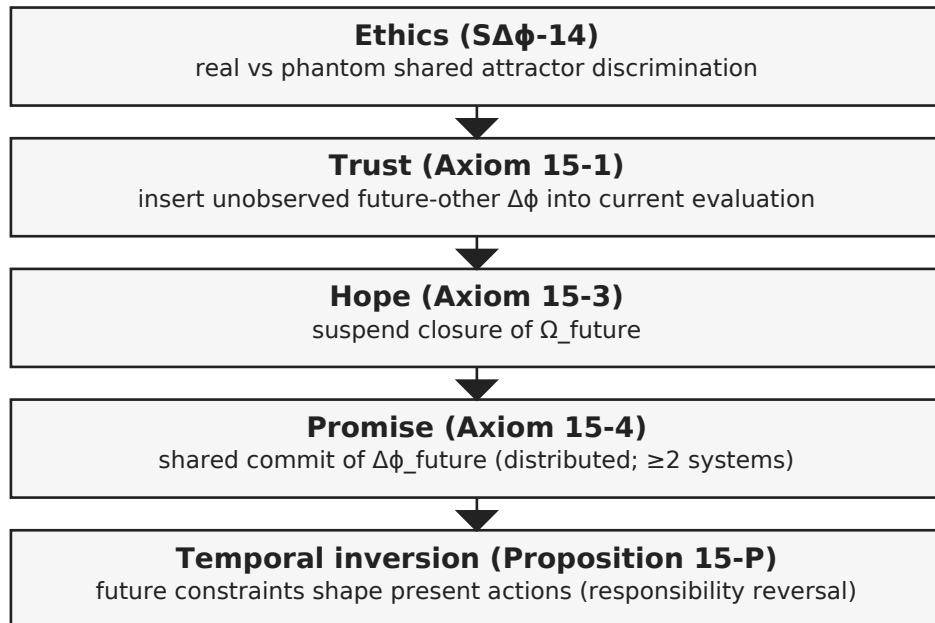
Sacred/ritual/institutional layers are excluded. This item fixes only minimal pre-commitment operators.

8. Exit question

If ethics validates relation reality, what counts as sufficient evidence to maintain a promise under uncertainty?

9. Core diagram — Pre-commitment stack

Minimal dependency map linking ethics ($S\Delta\phi$ -14) to pre-commitment operators ($S\Delta\phi$ -15).



Notes:

Without promise: trust remains fragile. Without trust: ethics cannot operate in time.

Note. The stack emphasizes temporal asymmetry: validation is delayed, therefore pre-commitment is necessary.